Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Simplification of algebraic expressions: order of operations, brackets and word problems. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Second Term
Week	9
Topic	Simplification of Algebraic Terms
Subtopic	Simplification of algebraic expressions: order of operations, brackets and word problems
Theme	—
Curriculum objectives	• carry out order of operation in algebra • insert/remove bracket and simplify expression • solve word problems involving algebraic expression
Previous knowledge	• Addition and subtraction of algebraic terms • Multiplication and division of algebraic terms
Lesson sequence	• carry out order of operation in algebra — the teacher solves some questions in order of operation on the board; the student listen, contribute, and where necessary ask questions where they do not understand; the student copy worked examples into their notes; the student solve some questions independently; the teacher moves round to monitor their work, answer their questions, mark their notes and mark correction. Set for the students: 1. $4n + 3n \times 10$ 2. $32 + 8x \div 2$ 3. $3 \times 5x + 4x \div 2$ 4. $36y \div 6 + y + 2 \times 4y$ • insert/remove bracket and simplify expression — the teacher solves some questions on insertion, removal of bracket on the board; the student listen, ask questions on their areas of difficulty; the student copy worked examples into their notes; the student solve some questions independently; the teacher moves round to monitor their work, answer their questions, marks their notes and makes corrections. Set for the students: (a) $(12a - 7) + (4 - 5a)$ (b) $(8x - 2y) - (9x - 4y)$ • solve word problems involving algebraic expression — the teacher solves some questions on word problems involving algebraic expression on the board; the student listen, ask questions where they do not understand; the student solve some questions independently; the teacher moves round to monitor their work, answer their questions, mark their notes and mark correction. Set for the students: 1. A pen cost h naira and a crayon cost ₦20 less than a pen. (a) Find the cost of a crayon (b) Find the cost of a crayon and a pen 2. Envelopes cost d naira per pack, 4 note pad cost 20 naira more than a pack of envelopes. Find the total cost of a pack of envelope and notepad.
Instructional materials	• Chart showing algebraic expression.
Evaluation	• Simplify (a) $14pq \times r$ (b) $12x^2y \div 3x$ • Simplify $7 \times 3a - (3a + 5a) \times 2$

	• Simplify $12-(7-2f)-(3-f)$ • Simplify $(8a-5)+(14-3a)$ • A pen and a pencil altogether cost x naira. If a pen cost y naira, find the cost of 3 pencils.

Lesson objectives

1.	carry out order of operation in algebra
2.	insert/remove bracket and simplify expression
3.	solve word problems involving algebraic expression

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Simplification of algebraic expressions: order of operations, brackets and word problems

Class: JSS1 **Subject:** Mathematics **Week:** 9

Section 1: Order of Operations in Algebraic Expressions

When an algebraic expression contains more than one operation, you cannot just work from left to right. You must carry out the operations in a fixed order, called the order of operations. Deal with anything inside brackets first, then work out divisions and multiplications (working left to right), and finally carry out additions and subtractions (again left to right).

Look at the expression $18 - 6 \times 2$. The multiplication must be done before the subtraction. So $6 \times 2 = 12$, and then $18 - 12 = 6$. If you subtracted first you would wrongly get $12 \times 2 = 24$.

The same rules hold when the terms contain letters. In $3x \times 2 + 5x$, do the multiplication first: $3x \times 2 = 6x$, then add the like terms: $6x + 5x = 11x$. When a term is divided, like $7x \div x$, write it as $\frac{7x}{x}$ and cancel: $\frac{7x}{x} = 7$.

Example 1 — find the value

(a) $18 - 6 \times 2$

Do the multiplication first: $18 - 12$

$= 6$

(b) $6 - 18 \div 3$

Do the division first: $6 - 6$

$= 0$

(c) $6 \times 2 - 2 - 40 \div 4$

Work the multiplication and division first: $(6 \times 2) - 2 - (40 \div 4)$

$= 12 - 2 - 10$

Now subtract from left to right: $10 - 10$

$= 0$

Example 2 — simplify

(a) $3x \times 2 + 5x$

Multiply first: $3x \times 2 = 6x$

$= 6x + 5x$

Add the like terms: $= 11x$

(b) $17y - 5 \times 2y$

Multiply first: $5 \times 2y = 10y$

$= 17y - 10y$

Subtract the like terms: $= 7y$

Example 3 — simplify

(a) $7x \div x + 5$

Work the division first: $7x \div x = \frac{7x}{x} = 7$

$= 7 + 5$

$= 12$

(b) $4x - (6 \div 3)$

Work inside the bracket first: $6 \div 3 = 2$

$= 4x - 2$

(c) $(24x \div 6) + x + (1 \times 5x)$

Work inside each bracket: $24x \div 6 = 4x$ and $1 \times 5x = 5x$

$= 4x + x + 5x$

Add the like terms: $= 5x + 5x$

$= 10x$

Practice

1. Simplify $4n + 3n \times 10$.
2. Simplify $32 + 8x \div 2$.
3. Simplify $3 \times 5x + 4x \div 2$.
4. Simplify $36y \div 6 + y + 2 \times 4y$.
5. Simplify $7 \times 3a - (3a + 5a) \times 2$.

Section 2: Removing Brackets and Simplifying Expressions

When an expression contains brackets, you remove them and then combine the like terms. Three hints will carry you through every example:

- Always try to simplify the terms inside the bracket first, if you can.
- A + sign before a bracket does not change the signs of the terms inside the bracket when the bracket is removed. Every term inside keeps its sign.
- A − sign before a bracket changes every + inside to − and every − inside to + when the bracket is removed.

Once the brackets are gone, you collect like terms exactly as you already know how to do for addition and subtraction of algebraic terms.

Example 1. Simplify (a) $a + (b + c)$, (b) $a + (b - c)$, (c) $2a - (3b - c)$.

(a) A + sign sits before the bracket, so every term inside keeps its sign:
$$a + (b + c) = a + b + c.$$

(b) Again a + sign sits before the bracket, so the − inside stays as −:
$$a + (b - c) = a + b - c.$$

(c) Here a − sign sits before the bracket, so the $3b$ stays, but the − before c turns into +:
$$2a - (3b - c) = 2a - 3b + c.$$

Example 2. Simplify (a) $a + (a - b)$, (b) $6x + (3 - x)$, (c) $(3a + 2) + (5a + a)$.

(a) The + before the bracket keeps every sign, so you write out the terms and then collect like terms:
$$a + (a - b) = a + a - b.$$
Combining the two a terms gives $2a - b$.

(b) Remove the bracket with the + in front:
$$6x + (3 - x) = 6x + 3 - x.$$
Collect the x terms together: $6x - x + 3$, which simplifies to $5x + 3$.

(c) Start by simplifying what is inside the bracket, since $5a$ and a are like terms: $5a + a = 6a$. So
$$(3a + 2) + (5a + a) = 3a + 2 + 6a.$$
Collect the a terms: $3a + 6a + 2$, which simplifies to $9a + 2$.

Example 3. Simplify (a) $5x - (3y + z)$, (b) $10d - (9c - 4d)$, (c) $(6x - y) - (4x - 5y)$.

(a) A − sign sits before the bracket, so every sign inside flips: the $3y$ keeps its + meaning as written, and the + before z turns into −:
$$5x - (3y + z) = 5x - 3y - z.$$
There are no like terms to combine, so this is already simplified.

(b) The − before the bracket flips the signs inside: $9c$ stays, and the − before $4d$ turns into +:
$$10d - (9c - 4d) = 10d - 9c + 4d.$$
Collect the d terms: $10d + 4d - 9c$, which simplifies to $14d - 9c$.

(c) Remove both brackets, watching the signs. The first bracket is added, so it keeps its signs: $6x - y$. The second bracket is subtracted, so both signs inside flip: $4x$ becomes $-4x$ and $-5y$ becomes $+5y$:

$(6x - y) - (4x - 5y) = 6x - y - 4x + 5y.$
Collect like terms: $6x - 4x = 2x$ and $5y - y = 4y$, giving $2x + 4y$.

Practice

1. Simplify $(12a - 7) + (4 - 5a)$.
2. Simplify $(8x - 2y) - (9x - 4y)$.
3. Simplify $(8a - 5) + (14 - 3a)$.
4. Simplify $12 - (7 - 2f) - (3 - f)$.

Section 3: Word Problems with Algebraic Expressions

Many algebra problems come to you written in words. Your job is to read the words carefully, decide which letter stands for which amount, and then write the situation as an algebraic expression before you simplify it.

Here are the two key moves.

Step 1 — pick a letter for each unknown. The problem usually tells you what a letter means. For example, "a pen cost ₦P" tells you P is the cost of a pen.

Step 2 — turn the words into an expression. Watch the little words that tell you to add or subtract:
- "15 naira less than a pen" means subtract 15 from the pen's cost: ₦(P − 15).
- "40 naira more than a pencil" means add 40 to the pencil's cost: ₦(x + 40).
- "5 boys join" means the number of boys increases by 5: x + 5.
- "3 girls leave" means the number of girls decreases by 3: y − 3.

Once the situation is written as an expression, simplify it the way you already know — add like terms and combine the numbers. Then put the ₦ sign back in front of the final answer.

Let's work through a few word problems step by step.

Example 1

A pen cost ₦P and a pencil cost 15 naira less than a pen.

(a) Find the cost of a pencil.
(b) Find the cost of a pen and a pencil.

Solution

The cost of a pen is ₦P.

(a) The pencil costs 15 naira less than the pen, so we subtract 15:
cost of a pencil = ₦(P − 15).

(b) A pen and a pencil together cost the pen plus the pencil:
= ₦P + ₦(P − 15)
= ₦P + ₦P − ₦15
= ₦2P − ₦15
= ₦(2P − 15).

So a pen and a pencil cost ₦(2P − 15).

Example 2

A pencil cost ₦x and a rubber cost 40 naira more than a pencil.

(a) Find the cost of a rubber.
(b) Find the cost of a pencil and a rubber.

Solution

The cost of a pencil is ₦x.

(a) The rubber costs 40 naira more than the pencil, so we add 40:
cost of a rubber = ₦(x + 40).

(b) A pencil and a rubber together:
= ₦x + ₦(x + 40)

= ₦(x + x + 40)
= ₦(2x + 40).

So a pencil and a rubber cost ₦(2x + 40).

Example 3

In a class, there are x boys and y girls last semester. This semester, 5 boys join and 3 girls leave the class. How many students are now in this class?

Solution

The total number of students is the number of boys plus the number of girls.

Number of boys = x + 5, because 5 boys join.
Number of girls = y − 3, because 3 girls leave.

Total = (x + 5) + (y − 3)
= x + 5 + y − 3
= x + y + 5 − 3
= x + y + 2.

So the class now has x + y + 2 students.

Practice

1. A pen costs ₦h and a crayon costs ₦20 less than a pen.
   (a) Find the cost of a crayon.
   (b) Find the cost of a crayon and a pen.
2. Envelopes cost ₦d per pack, and a notepad costs 20 naira more than a pack of envelopes. Find the total cost of a pack of envelopes and a notepad.
3. A pen and a pencil altogether cost ₦x. If a pen costs ₦y, find the cost of 3 pencils.

Check your work

Attempt every question before you read this page.

Order of Operations in Algebraic Expressions

1. $34n$

   Multiply first: $3n \times 10 = 30n$. Then $4n + 30n = 34n$.
2. $32 + 4x$

   Divide first: $8x \div 2 = 4x$. The 32 and $4x$ are not like terms, so the answer stays $32 + 4x$.
3. $17x$

   Work the multiplication and division first: $3 \times 5x = 15x$ and $4x \div 2 = 2x$. Then $15x + 2x = 17x$.
4. $15y$

   Work the division and multiplication first: $36y \div 6 = 6y$ and $2 \times 4y = 8y$. Then $6y + y + 8y = 15y$.
5. $5a$

   Work inside the bracket first: $3a + 5a = 8a$. Now the expression is $7 \times 3a - 8a \times 2$. Multiply: $7 \times 3a = 21a$ and $8a \times 2 = 16a$. Then $21a - 16a = 5a$.

Removing Brackets and Simplifying Expressions

1. $7a - 3$

   A + sits before each bracket, so keep all the signs: $12a - 7 + 4 - 5a$. Collect the a terms: $12a - 5a = 7a$, and the numbers: $-7 + 4 = -3$. So the answer is $7a - 3$.
2. $-x + 2y$

   The first bracket is added, so it keeps its signs: $8x - 2y$. The second bracket is subtracted, so both signs flip: $9x$ becomes $-9x$ and $-4y$ becomes $+4y$. So we have $8x - 2y - 9x + 4y$. Collect like terms: $8x - 9x = -x$ and $-2y + 4y = 2y$. The answer is $-x + 2y$.

3. $5a + 9$

   A + sits before each bracket, so keep all the signs: $8a - 5 + 14 - 3a$. Collect the a terms: $8a - 3a = 5a$, and the numbers: $-5 + 14 = 9$. So the answer is $5a + 9$.
4. $2 + 3f$ (or $3f + 2$)

   The first − before a bracket flips signs inside $7 - 2f$, giving $12 - 7 + 2f$. The second − before $3 - f$ flips both signs, giving $-3 + f$. Altogether: $12 - 7 + 2f - 3 + f$. Collect the numbers: $12 - 7 - 3 = 2$, and the f terms: $2f + f = 3f$. So the answer is $2 + 3f$, which is the same as $3f + 2$.

Word Problems with Algebraic Expressions

1. (a) ₦(h − 20)
   (b) ₦(2h − 20)

   (a) A crayon costs 20 naira less than the pen, so cost of crayon = ₦(h − 20).
   (b) Crayon and pen together = ₦(h − 20) + ₦h = ₦h − ₦20 + ₦h = ₦(2h − 20).
2. ₦(2d + 20)

   A pack of envelopes costs ₦d. The notepad costs 20 naira more, so it costs ₦(d + 20). Together: ₦d + ₦(d + 20) = ₦(d + d + 20) = ₦(2d + 20).
3. ₦3(x − y)

   Pen and pencil together cost ₦x, and the pen costs ₦y, so one pencil costs ₦(x − y). Three pencils cost 3 × ₦(x − y) = ₦3(x − y).

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-f057ef96	**Date**	
Subtopic	Simplification of algebraic expressions: order of operations, brackets and word problems	**Rater**	

The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____